

GY457

Applied Urban and Real Estate Economics

Revision Lecture

W11 Winter Term 2026

Wednesday 1 April 2025, 2-4.30pm, CBG.G.01

Exam Specifics, Preparation Advice and Q&A



Contents

1. Exam specifics
2. General Exam Preparation Advice
 - How to Prepare for Exam
 - During the Exam
3. Questions & Answers

Contents

1. Exam specifics
2. General Exam Preparation Advice
 - How to Prepare for Exam
 - During the Exam
3. Questions & Answers

Exam Specifics—Overview

- Exam is **in person**
- Two parts – **4 questions**
 - ▶ **Part 1** consists of 2 questions (Bagagli lectures & all AT seminars)
 - ▶ **Part 2** consists of 2 questions (Hilber lectures, Yu lecture & all WT seminars)
 - ▶ Each Part contains one **essay-style** question and one **partitioned** question
 - ▶ Questions may overlap topics
- **Answer 2 questions – one from each part**
- Time allowed: **3 hours** + 10 minutes reading time
- Calculator: Is not allowed (and not needed)

Example of Open-Ended Essay-Style Question

“Consumer mobility and inter-jurisdictional competition lead to an efficient provision of local public services.”

Critically discuss the validity of the above proposition by reference to the relevant literature that followed Charles Tiebout’s (1956, *Journal of Political Economy*) seminal paper. (**100 marks**)

(GY457 exam of 2012; Q4)

Example of a Partitioned Question

Estimating the housing supply and demand curves is an empirical challenge due to the so called ‘identification problem’.

- a) What exactly is the ‘identification problem’ in the context of estimating housing demand and supply curves? Explain. **(30 marks)**
- b) How have empirical researchers attempted to estimate the housing supply price elasticity? Discuss the underlying methodology and the main insights of the empirical literature. **(40 marks)**
- c) To what extent can it be helpful for investors and/or policy makers to know the housing supply price elasticity of a particular housing market? Discuss. **(30 marks)**

(GY457 exam of 2013/4; Q6)

Exam Specifics—Cont.

- No word limits – but longer answers not necessarily better
 - ▶ Not the quantity but the **quality** of answers is what matters
- Adding figures and equations is ok and sometimes useful
 - ▶ **Figures** can be particularly helpful but need to be well explained + axis etc. need to be labelled!
- Referencing
 - ▶ Do add **short references** (e.g., Tiebout (1956)) in text to refer to relevant articles, books etc. if appropriate
 - ▶ As this is an ‘in person’ exam there is no need for a reference section at end

Exam Specifics:

What makes a good/top answer? (1/2)

- If for some reason you cannot or do not want to use Digiexam, make sure your answer is legible
- Make sure answer is **focused** on question, yet answer is **comprehensive** within perimeter given by question
 - ▶ Do not just blindly reproduce lecture notes that only broadly relate to topic but do not answer what the question asks specifically
- Express your reasoning clearly; make sure your **line of reasoning is logical**

Exam Specifics:

What makes a good/top answer? (2/2)

- Answer should be **well structured, not just bullet points**
 - ▶ Structure particularly important for essay-style questions
 - ▶ Bullet points only in 'worst case', if run out of time
- Good/top answers are **thought-through & apply critical thinking**
 - ▶ Convey that you understood the mechanisms of a model, the empirical method employed, the chain of arguments invoked by authors etc.

Contents

1. Exam specifics
2. General Exam Preparation Advice
 - How to Prepare for Exam
 - During the Exam
3. Questions & Answers

How to Prepare for Exam?

- Individual differences
- Importance of
 - ▶ Lectures
 - ▶ Core readings and seminar readings (both compulsory!)
 - ▶ Seminar discussions
 - ▶ Understanding concepts, don't just memorise
 - ▶ Have time for repetition!

Preparations and Exam Itself...

- Get enough rest on night before exam
- Manage your time well!
 - ▶ If you have time: Do a (another) 'dry run' with a self-assembled mock exam selected from [past exam papers](#)
 - ▶ Allocate roughly equal time to questions (and proportional to marks for sub-questions)
 - ▶ Answer both questions fully!
- **Answer question and nothing else!!!**
 - ▶ Think carefully about what exactly question asks and don't just reproduce lecture notes

How to Answer Exam Questions?

- Distinguish between open **essay-style questions** and **partitioned questions**
- Open **essay-style questions**
 - ▶ Read question carefully and ‘break it down’ into elements
 - ▶ It is advisable to ‘sketch answer’ when mind is ‘fresh’ (structure, important elements)
 - ▶ Structure your answer well with (i) intro, (ii) main body/analysis (e.g., pros and cons) & (iii) concluding section
 - ▶ Make sure there is a logical thread throughout your essay (sections build on and logically relate to each other)
 - ▶ Give answer that reflects all the main aspects of the problem/issue/topic (but nothing else!)

How to Answer Exam Questions?—Cont.

■ Partitioned questions

- ▶ Answer sub-question ‘to the point’ (be concise)
 - No need for intro & conclusion
 - But be comprehensive within perimeter given by sub-question
- ▶ Don’t lose time with answering one sub-question in excessive detail leaving no time for rest
 - Weight per sub-question will be noted (as number of marks out of a 100 marks)
 - If nothing is noted, assume equal weight

What to Do if Things Do Not Go as Expected?

- The hurdle race analogy
 - ▶ If it is difficult for you, it is likely difficult for the others as well
 - ▶ Stay calm!

Contents

1. Exam Specifics
2. General Exam Preparation Advice
 - How to Prepare for Exam
 - During the Exam
3. Questions & Answers

Lastly...



Best of luck
with the
exams!!!



Mock exam Q3 (Part 2) → Topic 2

- **Essay style question:**

In a 2013 Journal of Property Research article, Paul Cheshire argued:

“Planning serves valuable ends. Land market failure is potentially a serious problem. We need a strategic mechanism for coordinating land uses and infrastructure provision. The evidence very strongly indicates, however, that in Britain at least, problems of policy failure are greater than those of market failure. We could secure most of the benefits that planning generates at much lower cost to society. Planning needs to work with as well as against markets.”

Explain the meanings of “market failure” and “policy failure” in the context of land use regulation, and critically discuss this statement with reference to the relevant theoretical and empirical literature. (100 marks)

Analyzing the question

- Cheshire (2013) argues planning has **valuable purposes** (welfare economic justification is correcting for market failure), but
- In Britain: **policy failure > market failure**
- Claim: same benefits could be achieved at **lower social cost**
- Key implication: planning should **work with markets, not against them**
- **Task:**
 - ▶ Define **market failure** and **policy failure**
 - ▶ Critically assess the claim **building on theory + using evidence**

General points

- A good answer is **well structured**, this entails
 - ▶ An **introduction**
 - Revisits and motivates the question + explains structure of essay
 - ▶ A **main body** of the essay
 - Provides the core of the analysis (refers back to question)
 - ▶ A **synthesizing conclusion** section
 - The conclusion ideally does not just repeat the analysis but really derives some conclusions based on the analysis

Defining market and policy failure (1/2)

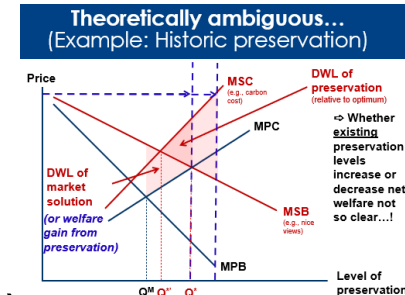
- Strong answer begins by carefully unpacking concepts of **market failure** and **policy failure** within context of LUR (i.e., welfare economics view vs. political economy view)
- **Market failure** refers to situations in which unregulated land markets fail to deliver efficient outcomes, due to
 - ▶ Pos. **externalities** (e.g., nice views on protected landmark buildings)
 - ▶ Neg. externalities (such as congestion or environm. degradation)
 - ▶ Under-provision of **public goods** (e.g., historic building, public park)
 - ▶ Coordination failures (**collective action problem** & strategic hold out problem) – Examples: Land assembly for (re-)development; new infrastructure provision

Defining market and policy failure (2/2)

- **Policy failure** refers to inefficiencies introduced by government intervention—particularly when planning regulations distort market signals, restrict housing supply excessively, or allocate land in a manner misaligned with demand
 - ▶ Policy failure can arise because decisions are not taken by benevolent planner but are **outcome of political process / vested interests** (not necessarily efficient)
 - ▶ Plus: **Information problem**: Difficulty to quantify extend of corrective actions – e.g., how much preservation in a city is optimal

Critical assessment (main analysis)

- Good starting point could be to discuss the main theories
 - ▶ Welfare economics view (possibly with graph illustrating difference b/w 'market solution' and 'social optimum')
 - ▶ Political economy view (voting + lobbying): Homevoter hypothesis [Fischel 2000] + influential landowner hypothesis [H & R-N 2013] (+ ideology)
- Essay could then argue that whether $MB > MC$ at chosen regulatory stringency is ultimately an empirical question
 - ▶ Theory alone cannot answer
 - ▶ LUR has potential to correct market failure (MB)
 - ▶ But political process & vested interests may ultimately lead to policy failures (MC) that outweigh benefits



Empirical evidence (1/3)

- Could start with discussion of studies that focus on quantifying **MB** of LUR
 - ▶ Literature on preservation policies (e.g., Been et al. 2016, Koster et al. 2016, Ahlfeldt 2017)
 - ▶ Some evidence of capitalization of positive spillover effects in house prices
 - ▶ But be careful with interpretation
 - MC & MB are not always fully capitalized (carbon footprint; existence value)
 - Capitalization merely suggests preservation benefits marginal homebuyer, not necessarily welfare maximizing!
 - DiD does not capture aggregate 'market-wide' effects (constraints on aggregate supply driving up house prices across the entire metro area)

Empirical evidence (2/3)

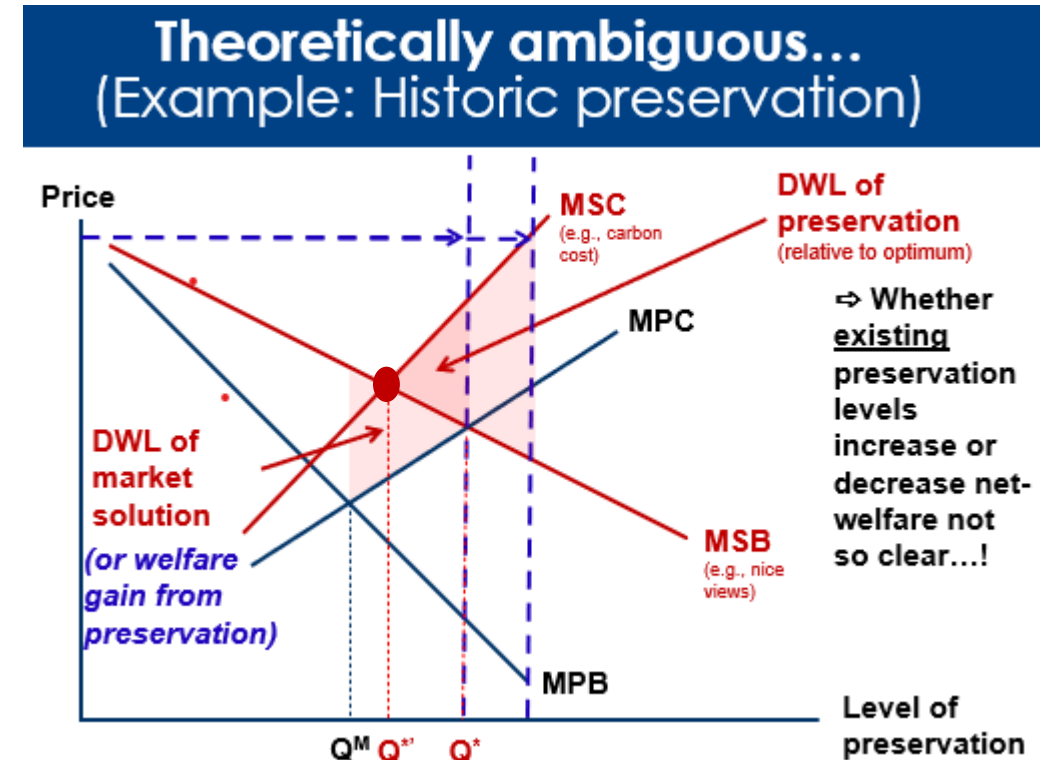
- Could then discuss long list of studies that suggest LUR is associated with important **costs**
 - ▶ LUR / NIMBYism drive up **cost of housing & commercial space** → RT [Glaeser et al. 2005; Cheshire & Hilber 2008; see also Hilber & Vermeulen 2016; Greaney et al. 2025]
 - ▶ LUR increase **vacancies & commuting distances** [Cheshire et al. 2018]
 - ▶ LUR has **adverse effects on output & productivity** → RT [Hsieh & Moretti 2019; Duranton & Puga 2023; Cheshire et al. 2015]
 - ▶ Preservation **increases energy consumption + social carbon cost** [Hilber et al. 2019]

Empirical evidence (3/3)

- What about **net welfare** effect?
 - ▶ Difficult to quantify & studies that focus on house prices are imperfect
 - ▶ But a few studies have attempted to assess [Turner et al. 2014; Cheshire and Sheppard 2002]
 - Suggestive that LUR in U.S. & in UK **too restrictive**

Conclusion / synthesis

- Empirical evidence seems to support claim of Cheshire in that LUR far from 'social optimum'
 - ▶ With optimum defined as $MSC=MSB$ ●
 - ▶ Political outcome is to the right of optimum
- DWL of (excessive) regulation may be greater than DWL of market solution



How was Q3 answered? (1/3)

- **Important caveat:** Snapshot in W8 without preparation...
- That said...
 - ▶ Some students did not answer any question from Part 2
 - ▶ Some students answered both questions (!) (but only very briefly)
 - ▶ Answers are typically one or two paragraphs or a few bullet points rather than a proper essay – very few exceptions
 - ▶ Some answered Part 1, but then ran out of steam for Part 2
 - ▶ Some just tried out the Digiexam system
 - ▶ Learning at this point is very superficial, suggesting focus on other courses(!)...

How was Q3 answered? (2/3)

- Most students could loosely distinguish between market failure & policy failure
 - ▶ But mostly only at a superficial level; very vague ; or incorrect (e.g., confusing shocks with failure)
 - ▶ Many students did not explicitly discuss externalities, public goods, collective action problem
 - ▶ Most students did not discuss political economy theories (homevoter hypothesis, influential landowner hypothesis) → first order conceptual gap!
- Very few students engaged with actual claim in question (i.e., policy failure > market failure)

Analysis of answer (Q3) (3/3)

- Minimal use of empirical literature
 - ▶ Evidence discussed often anecdotal or incorrect
 - Often weak structure and synthesis
 - ▶ No clear introduction
 - ▶ Conclusions often restate rather than synthesize
 - Overall diagnosis
 - ▶ Most students have basic intuition
 - ▶ But lack conceptual precision + theoretical depth + in depth knowledge of literature
- ⇒ Answers resemble superficial lecture recall or informal discussion, not yet analytical essays (with very few exceptions)

Mock exam Q4 (Part 2) → Topic 3 (& 5)

■ Partitioned question:

If you were tasked to estimate the supply price elasticity for your home region (or any other region with which you are familiar and for which there is available data), how would you go about it? Answer the following sub-questions.

- a. What empirical difficulties would you need to overcome? Discuss. (30 marks)
- b. What empirical approach/methodology would you choose and why? (Explain the empirical approach/method briefly) (20 marks)
- c. What would your estimating equation or equations look like, exactly? Discuss the rationale for the included variables, functional form etc. (20 marks)
- d. What are the locational factors that are likely to determine the supply price elasticity? Discuss with reference to the relevant literature. (30 marks)

General points

- Answers **focus** on exactly what the sub-questions ask
 - ▶ No need for introduction; get straight to the point
- Yet answers are **comprehensive** within perimeter given by question

What makes a good answer? Part a)

- Empirical difficulties
 - ▶ Main difficulty: Identification problem (explain)
 - ▶ Lack of good data
 - ▶ Specifying correct estimating equation; functional form
 - ▶ Validity of instruments (here: demand shifter; could refer to Bartik instrument)
 - ▶ Strength of identification (power of instrument)
 - ▶ Etc.

What makes a good answer? Part b)

- Method commonly used & discussed: **IV/TSLS**
 - ▶ Explain idea of IV: Need demand shifter (ideally Bartik-shift/share measure) to identify supply price elasticity
 - ▶ Explain two stages of TSLS
 - First stage: Regress ΔP_t (endogenous RHV) on ΔZ_t (excluded instrument) $\rightarrow \Delta \hat{P}_t$
 - Second stage: Regress ΔQ_t on $\Delta \hat{P}_t \rightarrow$ unbiased estimate of inverse of slope of supply curve
 - ▶ Why does this help? Crucial assumption is: Z only affects demand (not supply) $\rightarrow$ Idea: Keep S fixed, move D , get slope of S (linear approximation)

What makes a good answer? Part c)

- We are tasked to estimate supply price elasticity

Supply Equation:

$$\Delta Q_t^S = \alpha_0 + \alpha_1 \Delta \hat{P}_t + \alpha_2 (\Delta \text{Supply Shifters}_t) + \varepsilon_t \quad (4.1)$$

Demand Equation:

$$\Delta Q_t^D = \beta_0 + \beta_1 \Delta \hat{P}_t + \beta_2 (\Delta \text{Demand Shifters}_t) + \mu_t \quad (4.2)$$

Predicted value from 1st stage:
Estimated using instrumental
variables (=predetermined supply
and demand shifters)

- Important points

- ▶ Regress changes on changes – ΔP on ΔQ (construction) or $\partial \ln P_j \approx \Delta P/P$ & $\partial \ln H_j \approx \Delta Q/Q$ (explain why → all stationary)
- ▶ Explain importance of demand shifter (Bartik – how constructed) as instrument (used to predict change in P)
- ▶ Any supply shifters as included controls?
- ▶ Provide rationale for included variables

What makes a good answer? Part d)

- We discussed role of physical, topographical & regulatory supply constraints
 - ▶ Could discuss Hilber & Vermeulen (2016) for Engl. or Saiz (2010) for US
 - ▶ Could go in quite a bit of detail on HV (2016) as key reading discussed prominently in Topic 5
 - How are the constraints measured...
 - What are the findings of literature (relative importance)
 - Could even discuss endogeneity issues of the supply constraints measures and how they were resolved
 - ▶ Could also explain why less developable land makes supply inelastic (more costly to build, political economy argument, real options argument)

How was Q4 answered? (1/3)

- **Same caveat:** Snapshot in W8 without preparation...
- Same picture as for Q3
 - ▶ Learning at this point is very superficial, suggesting focus on other courses(!)

How was Q4 answered? (2/3)

- Most students (who attempted question in earnest) understood key empirical difficulty: endogeneity / identification problem
- Several students correctly recognised that prices and quantities are jointly determined & that there is a need for instruments (IV/TSLS)
- Some students recognized need of demand shifter, some discuss Bartik shift-share measure (but no detail)
- Broad idea of relevant supply constraints
- Some awareness of data challenges

Analysis of answer (Q4) (3/3)

- What majority of students systematically missed
 - ▶ Lack of clarity on identification strategy: rarely explain how identification works
 - ▶ Few explain D-shifters identify supply curve (and vice versa)
 - ▶ A few suggested DiD instead of IV/TSLS(!)
 - ▶ Part c) poorly answered, equation mostly incorrect, no clear specification in changes (as per Mayer & Somerville 2000)
 - ▶ No discussion of functional form issues, stationarity etc.
 - ▶ Mix up of what is being estimated
 - ▶ Missing or underdeveloped: instrument validity, weak instruments
 - ▶ Literature engagement is very limited
 - ▶ Answers generally very incomplete

⇒ **The “good news” is...**

The good news is...

- **All of this is fixable!**
 - ▶ With focused preparation & serious efforts over coming weeks
- **But** focus now needs to shift to GY457!

Lastly...



Best of luck with the exams!!!

